

Docker & Git Commands Reference Sheet

Cloud Computing Course

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Linux Commands

pwd

Prints the current working directory.

ls

Lists files and folders in the current directory.

cd <directory>

Changes the current directory.

mkdir <folder>

Creates a new folder.

touch <file>

Creates an empty file.

cat <file>

Displays the contents of a file.

echo "text"

Prints text to the terminal.

echo "text" > file.txt

Writes text into a file.

nano <file>

Opens a file in the Nano text editor.

Git Commands

git clone <repository-url>

Downloads a GitHub repository to your local machine.

git fetch

Downloads latest branch information from GitHub without merging changes.

git branch

Displays available branches.

git checkout <branch>

Switches to another branch.

git checkout -b <branch>

Creates a new branch and switches to it.

-b

Create a new branch before switching.

git status

Shows modified, staged and untracked files.

git add .

Stages all modified files for commit.

git commit -m "message"

Creates a commit with a message.

-m

Allows commit message to be provided inline.

git pull origin main

Downloads and merges latest changes from GitHub.

git push origin main

Uploads local commits to GitHub.

Docker Installation Commands

sudo dnf update -y

Updates installed packages.

-y

Automatically answers "yes" to prompts.

sudo dnf install docker -y

Installs Docker.

sudo systemctl start docker

Starts Docker service.

sudo systemctl enable docker

Starts Docker automatically after reboot.

sudo systemctl status docker

Displays Docker service status.

docker --version

Shows installed Docker version.

sudo usermod -aG docker ec2-user

Adds the user to the Docker group.

-a

Append user to existing groups.

-G

Specify supplementary group.

newgrp docker

Refreshes group membership without logout.

Docker Image Commands

docker images

Lists locally available Docker images.

docker pull nginx

Downloads an image from Docker Hub.

docker pull hello-world

Downloads the Hello World image.

Docker Container Commands

docker run hello-world

Creates and runs a Hello World container.

docker run -d -p 80:80 nginx

Runs Nginx container in background and maps port 80.

-d

Detached mode (run in background).

-p

Port mapping.

80:80

Host Port : Container Port

docker run -d -p 8080:80 nginx

Maps host port 8080 to container port 80.

docker ps

Lists running containers.

docker ps -a

Lists all containers including stopped containers.

-a

Show all containers.

docker stop <container-id>

Stops a running container.

docker start <container-id>

Starts a stopped container.

docker rm <container-id>

Deletes a container.

docker rmi <image>

Deletes a Docker image.

Interactive Containers

docker run -it ubuntu bash

Starts an Ubuntu container with an interactive terminal.

-i

Interactive mode.

-t

Allocates a terminal.

bash

Starts Bash shell inside container.

exit

Exits the container shell.

Docker Volumes

docker volume create mydata

Creates a Docker volume.

docker volume ls

Lists Docker volumes.

docker volume inspect mydata

Displays volume details.

docker run -it -v mydata:/data ubuntu bash

Starts container with volume attached.

-v

Mounts a volume.

mydata:/data

Volume Name : Container Path

cat /data/test.txt

Displays file stored in volume.

Dockerfile Commands

FROM

Specifies the base image.

Example:

FROM ubuntu

COPY

Copies files from host into image during build.

Example:

COPY index.html /usr/share/nginx/html

CMD

Specifies the default command executed when container starts.

Example:

CMD ["echo","Hello World"]

Docker Build Commands

docker build -t myimage .

Builds an image from Dockerfile.

build

Creates image from Dockerfile.

-t

Assigns a name/tag to image.

.

Current directory containing Dockerfile.

docker run myimage

Runs container from custom image.

Networking Concepts

Port Mapping

Connects host ports to container ports.

Example:

8080:80

Meaning:

Host Port 8080 ↓ Container Port 80

Security Group

AWS virtual firewall controlling inbound and outbound traffic.

Common Ports

22 → SSH

80 → HTTP

443 → HTTPS

3306 → MySQL

5432 → PostgreSQL

8080 → Alternate Web Application Port

Docker Workflow Summary

Dockerfile ↓ docker build ↓ Image ↓ docker run ↓ Container

Git Workflow Summary

git clone ↓ Modify Files ↓ git add ↓ git commit ↓ git push ↓ GitHub Repository

CI/CD Workflow Summary

Developer ↓ Git Push ↓ GitHub Actions ↓ Build ↓ Deploy ↓ Cloud